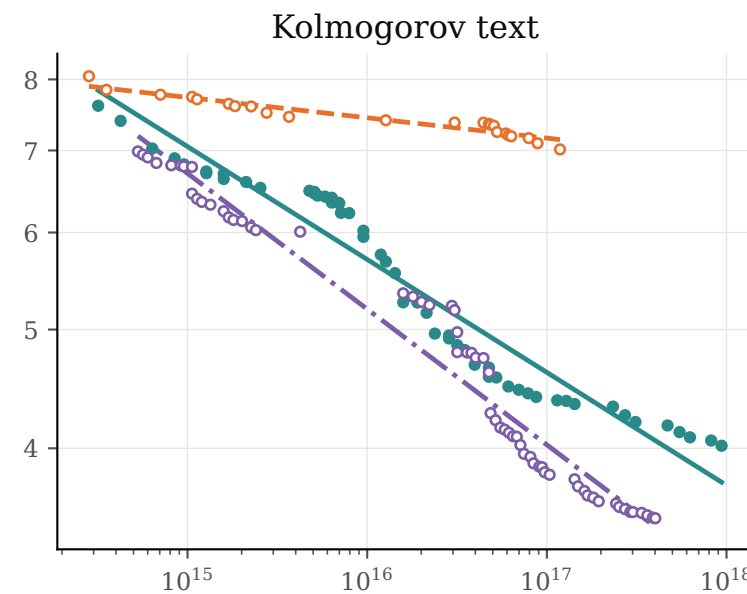
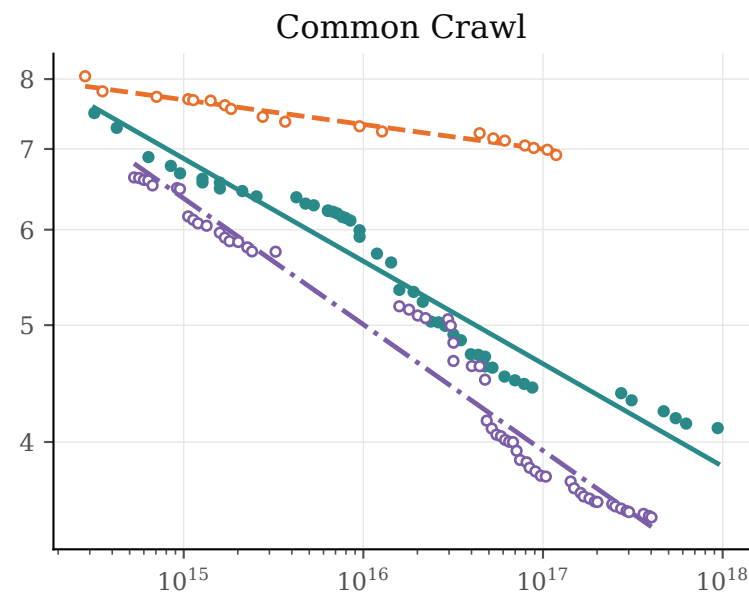
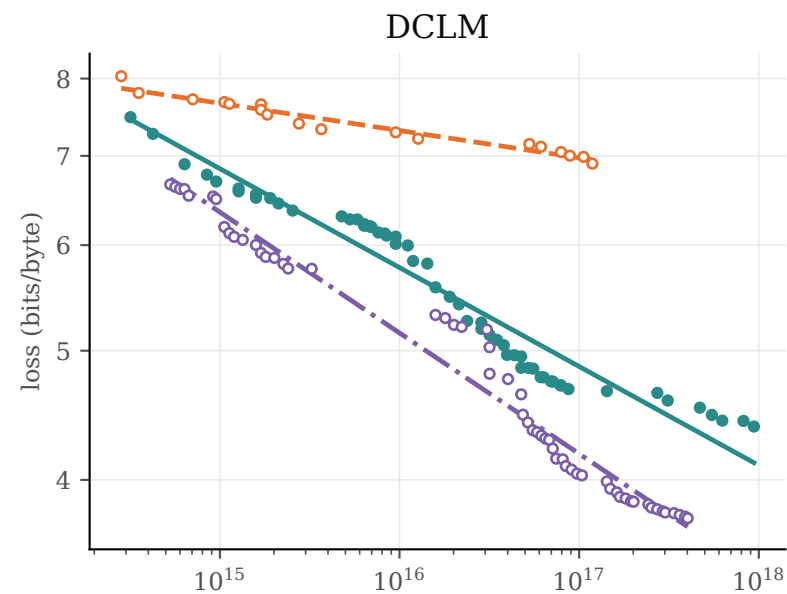


Self-Play Pretraining
with Zero Data

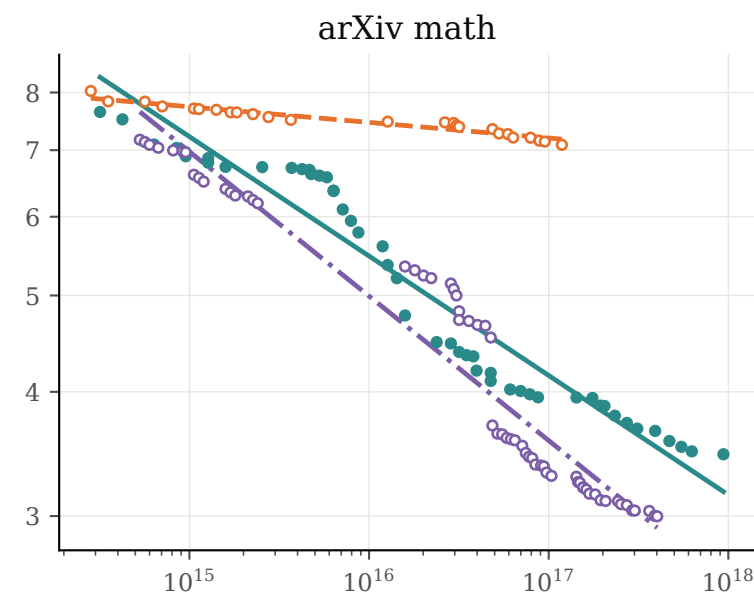
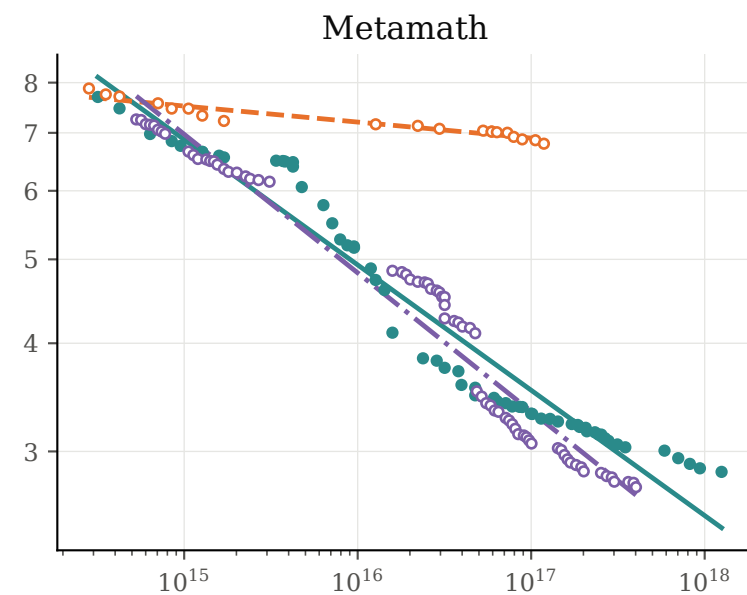
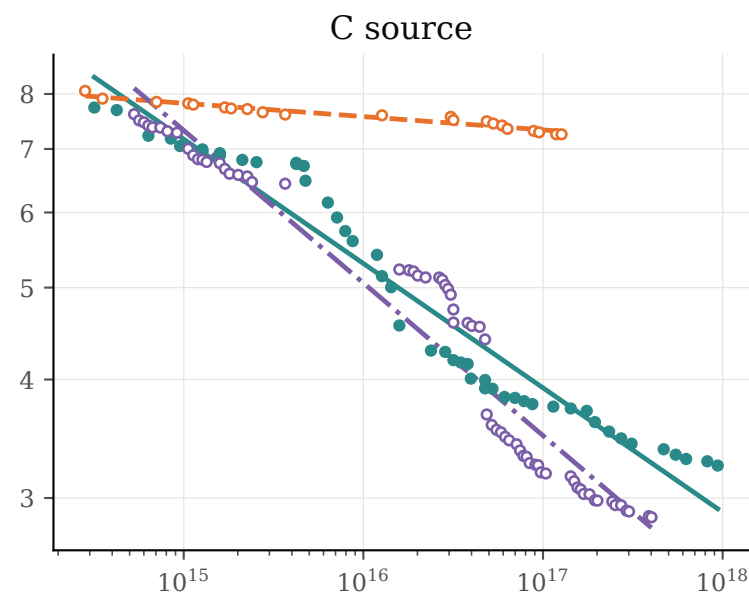
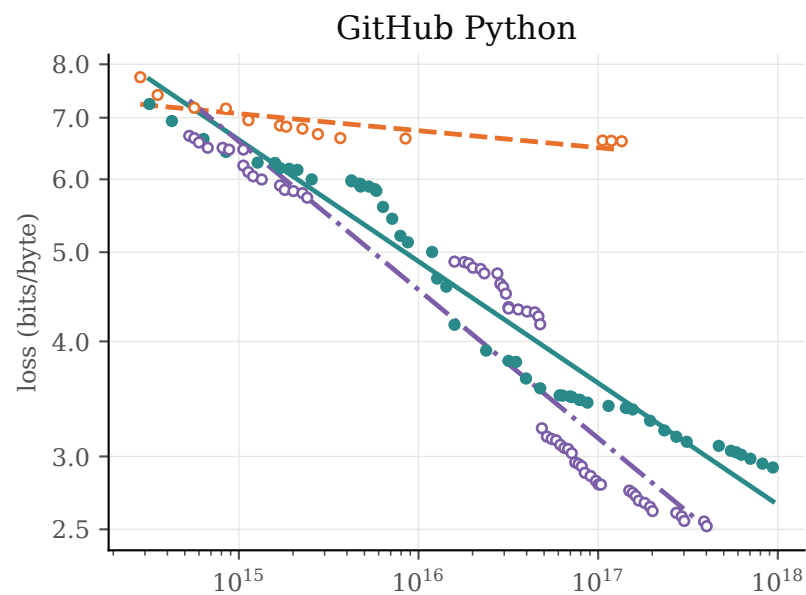
Pretraining on programs sampled
from a universal prior

Pretrain on PCFG

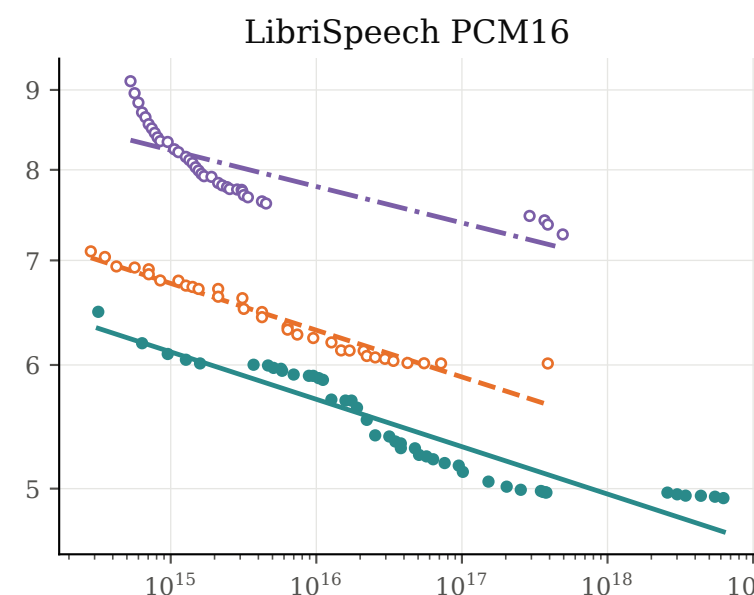
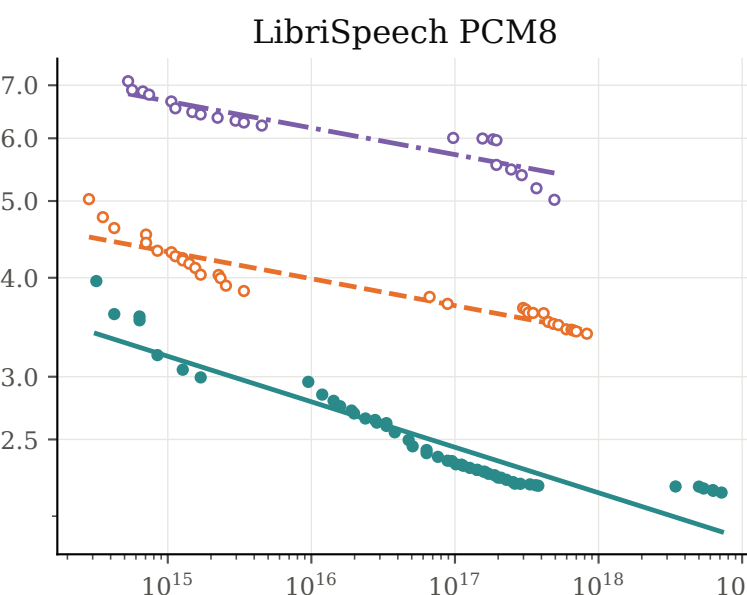
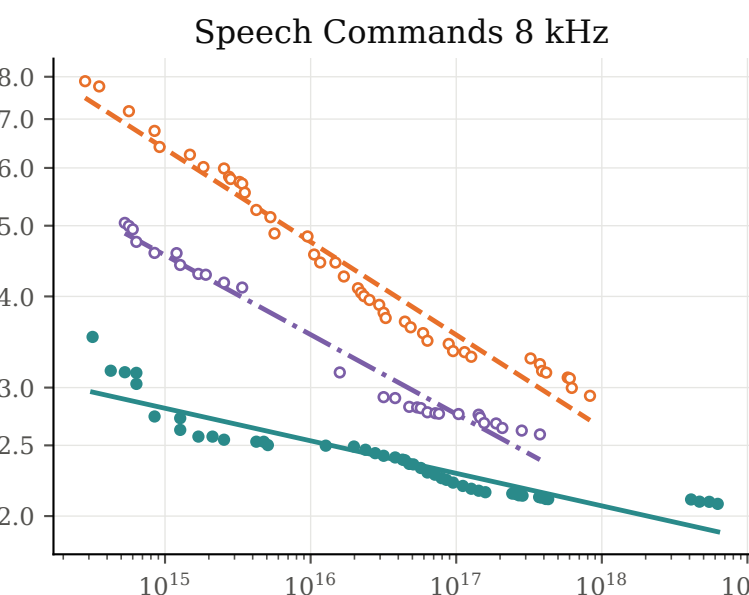
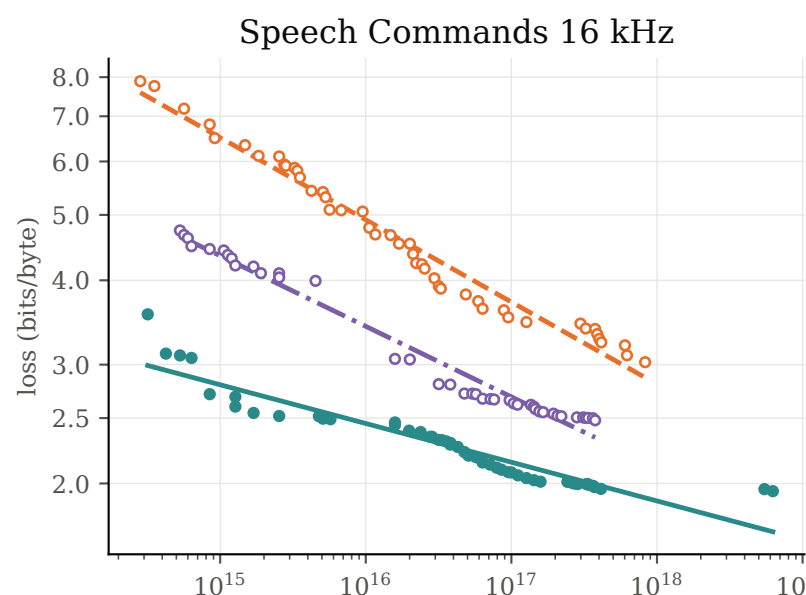
Text



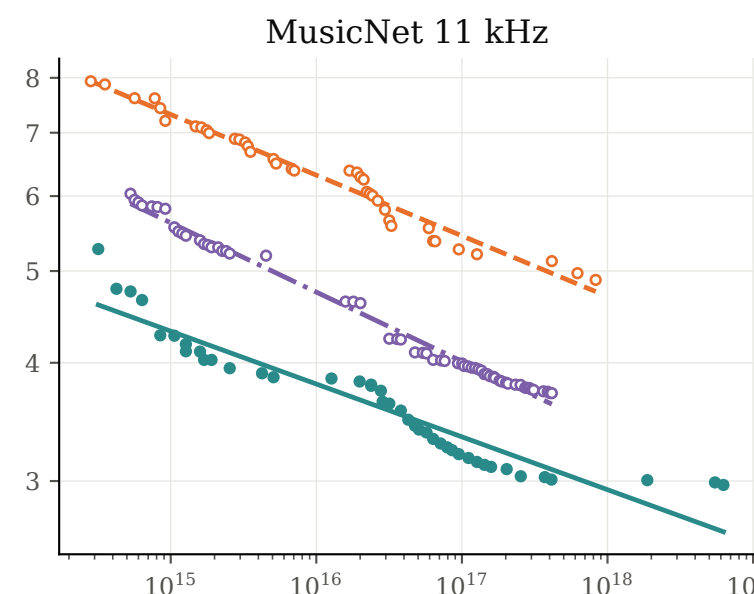
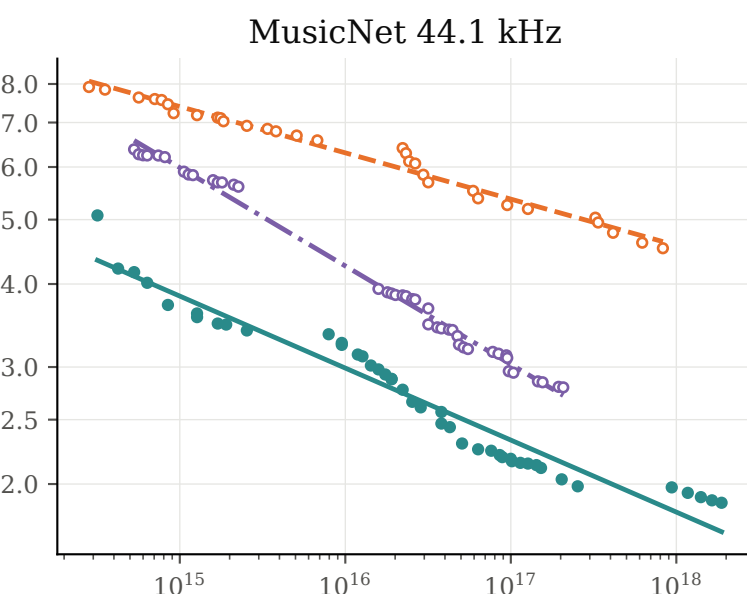
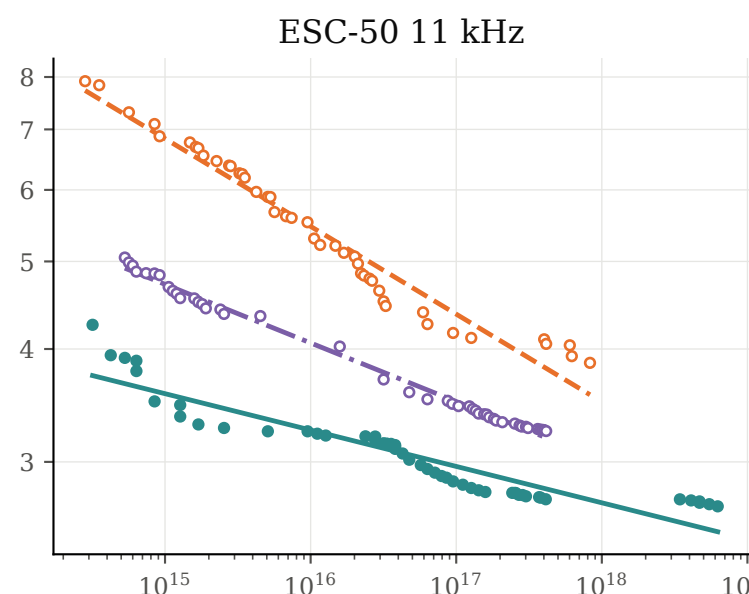
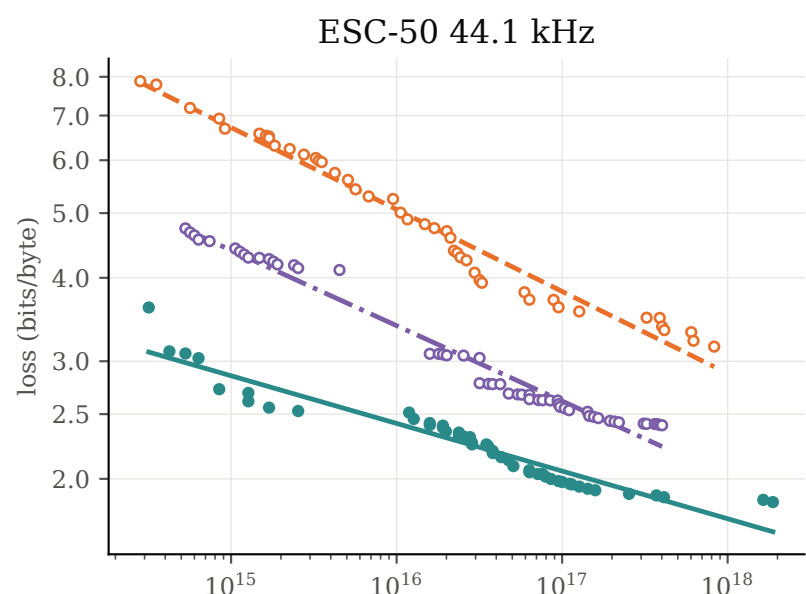
Code & formal



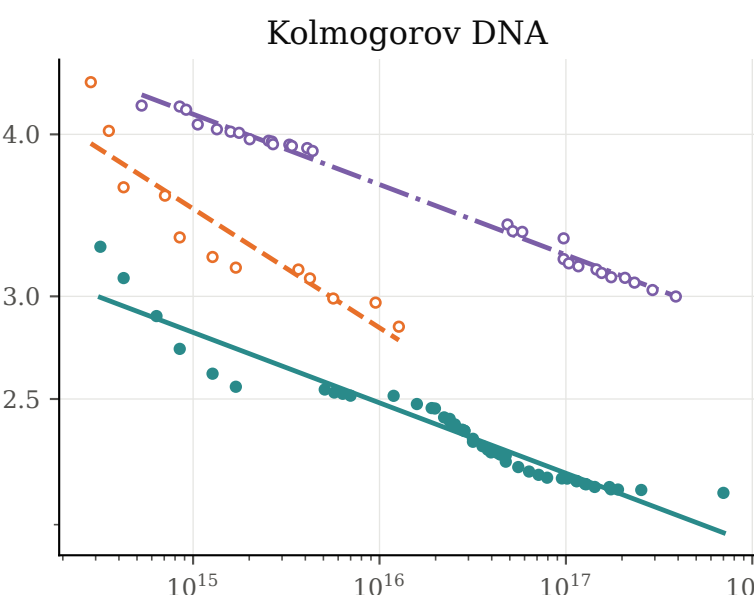
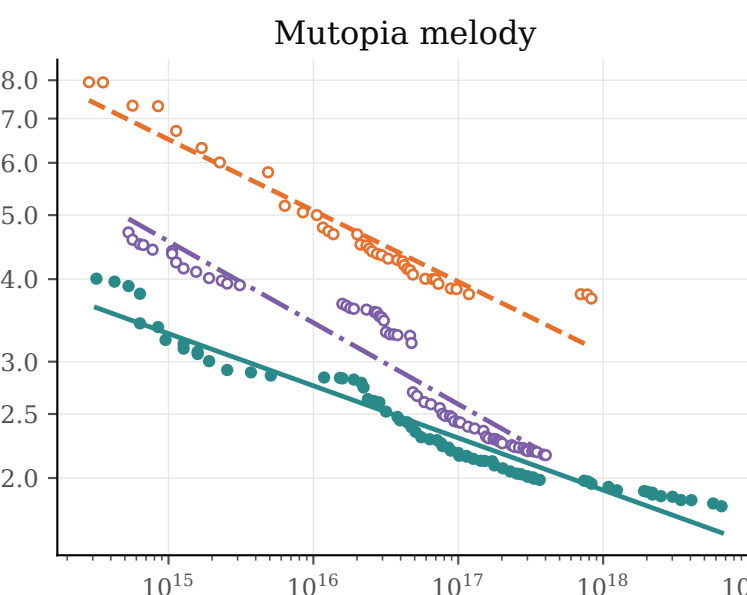
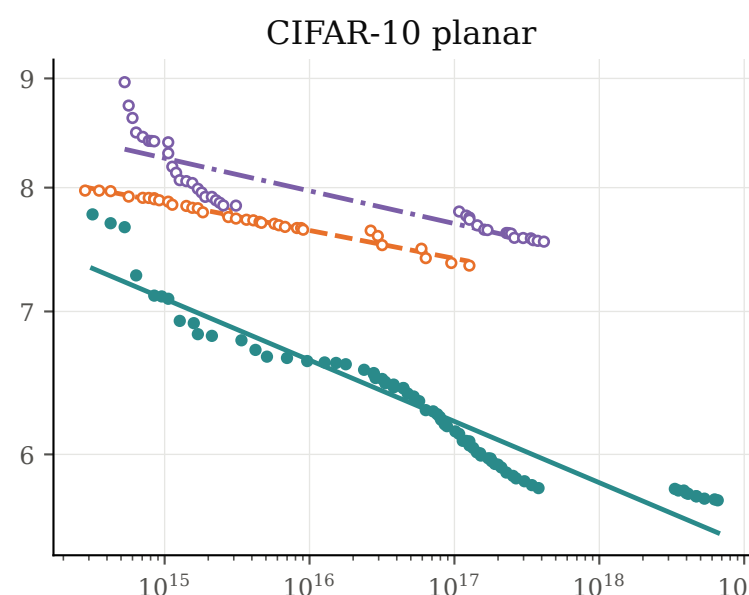
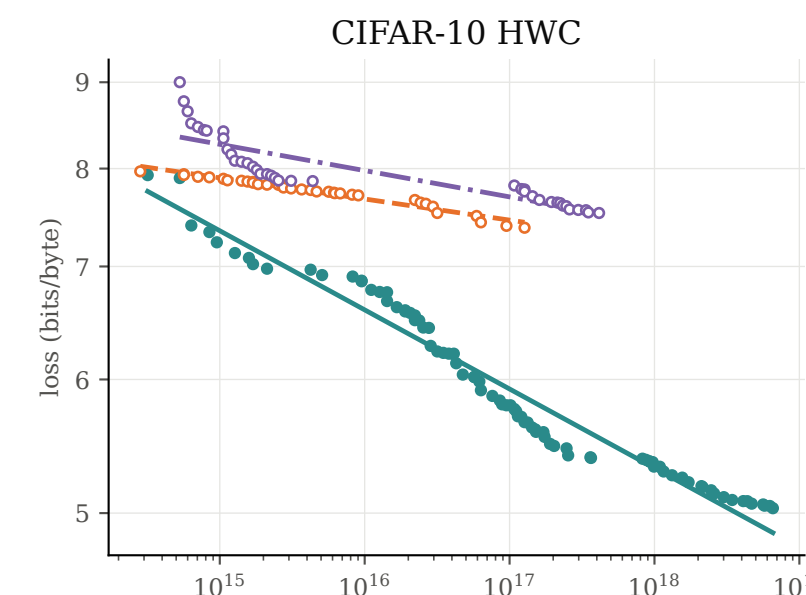
Speech audio



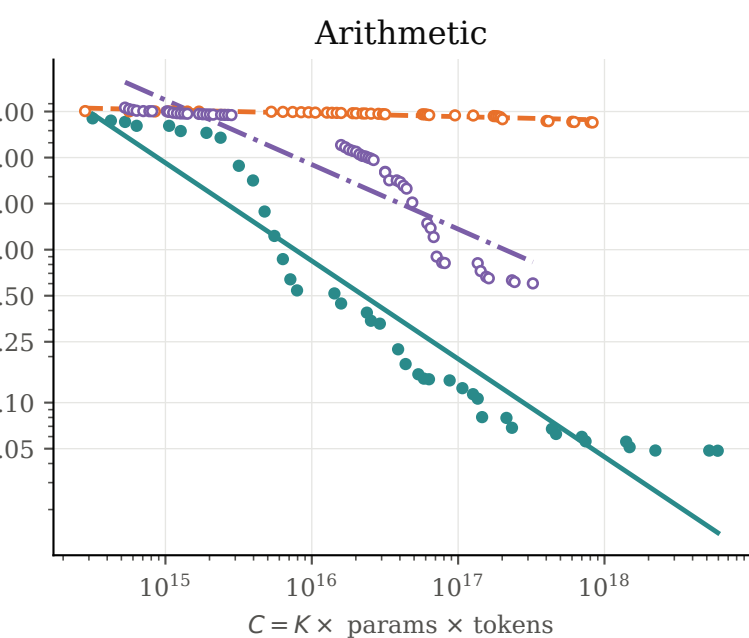
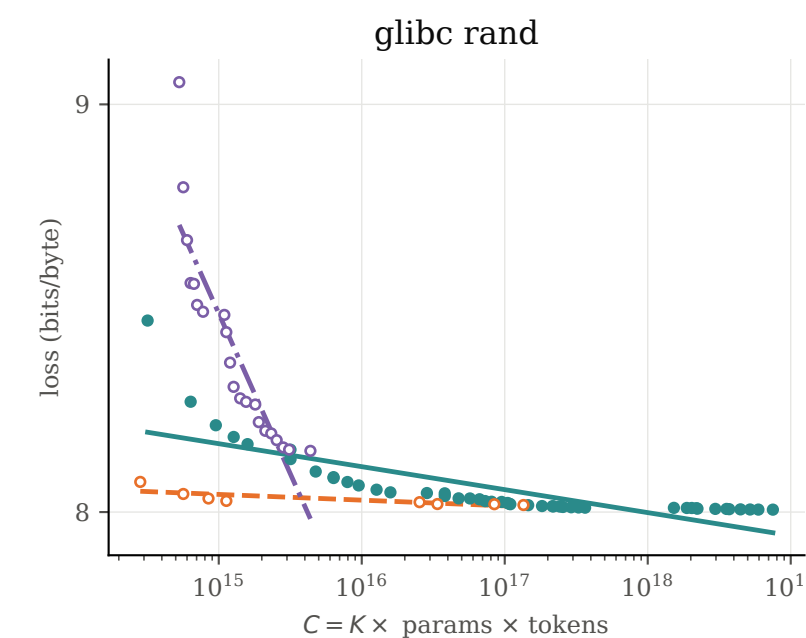
Sound & music audio



Vision, melody & DNA



Binary, numeric & synthetic



$C = K \times \text{params} \times \text{tokens}$

$C = K \times \text{params} \times \text{tokens}$